EE2005: Electrical Machines
Tutorial 8

1. Consider a stator coil of N turns, with the sides of the coil separated 180° apart. The length of the coil side is l. The coil links a flux density waveform produced by the rotor which varies as $B(\phi) = B_m \cos(\phi)$, where ϕ is the angle around the circumference of the stator. With respect to this waveform, the coil sides may be taken to be at angles $\phi = 0°$ and $\phi = 180°$ and the rotor field axis positioned at $\phi = 0^o$. With respect to this waveform, the coil sides may be taken to be at angles $\phi = 0^o$ and $\phi = 180^o$.

   (a) If this flux density waveform now rotates in the air gap due to the rotation of the rotor, find the expression for B when the rotor is at some angle $\theta = \omega\, t$.

   (b) Derive an expression for the flux linkage of the stator coil at any angle θ of the rotor. (Hint: Find the flux passing through an elemental area at an angle ϕ, and integrate to find the total flux passing through the area spanned by the coil).

2. Now consider another identical coil placed in the adjacent slot, which is situated at an angle of δ away from the first, i.e the coil sides may be assumed to lie at angles $\phi = \delta$ and $\phi = \pi + \delta$. Derive the expression for the voltage induced in that coil. If the two coils are now connected in series, what would be the net voltage?

3. A 208 V, 10 hp, 4 pole, 60 Hz, Y-connected induction motor has a full-load slip of 5 %.

   (a) What is the synchronous speed of this motor?

   (b) What is the rotor speed of this motor at the rated load?

   (c) What is the rotor frequency of this motor at the rated load?

   (d) What is the shaft torque of this motor at the rated load?

4. A 460 V, 25 hp, 60 Hz, 4-pole, Y-connected induction motor has the following impedances in ohms per phase referred to the stator circuit:
   $R_1 = 0.641\,\Omega$, $R_2' = 0.332\,\Omega$, $X_1 = 1.106\,\Omega$, $X_2' = 0.464\,\Omega$, $X_m = 26.3\,\Omega$
   The total rotational losses are 1100 W and are assumed to be constant. The core loss is lumped in with the rotational losses. For a rotor slip of 2.2% at the rated voltage and rated frequency, find the motor's

   (a) Speed

   (b) Stator current

   (c) Power factor

   (d) Resistance equivalent to mechanical load

5. The following results were obtained on a three phase, 75 kW, 3.3 kV, 6-pole, 50 Hz, Y-connected squirrel cage induction motor.
   No load Test: $f = 50$ Hz, $V_0 = 3.3$ kV, $I_0 = 5$ A, $P_0 = 2500$ W

Blocked rotor Test: $f = 15\,\text{Hz}$, $V_{BR} = 400\,\text{V}$, $I_{BR} = 27\,\text{A}$, $P_{BR} = 15\,\text{kW}$
The stator resistance per phase is $3.75\,\Omega$.

(a) Determine the parameters of the circuit model.

(b) For a slip of 0.04, calculate the stator current, its power factor and motor efficiency.